

Total No. of Questions : 4]

SEAT No. :

PC-340

[Total No. of Page : 1

[6361] - 227

B.E. (Computer Engineering) (In Sem.)
Honours in Artificial Intelligence & Machine Learning
Machine Learning
(2019 Pattern) (Semester - VII) (410301)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Answer Q.1 or Q.2, Q.3 or Q.4.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Assume suitable data, if necessary.*

- Q1)** a) Define Machine Learning and state applications of Machine Learning. [5]
b) State and explain types of learning in detail. [5]
c) What is mean by Probably Approximately Correct (PAC) learning? Write the use with example. [5]

OR

- Q2)** a) What is the Least Squares Regression method and why it is used? [5]
b) What is the difference between PAC and Factor Analysis? [5]
c) State and discuss the difference between linear regression and Logistic Regression. [5]

- Q3)** a) Write the purpose of Shrinkage Methods of Linear Regression. [5]
b) Why is Dimensionality Reduction important in Machine Learning. [5]
c) Short note on: Quadratic approximations and inference. [5]

OR

- Q4)** a) Define Linear Discriminant Analysis (LDA) Explain with example. [5]
b) What do you mean by a linear regression which applications are best modeled by linear regression? [5]
c) Short note on: Isomap [5]

